

Skills Summary

Reading and Plotting Points on a Grid

Use this reference while completing your worksheet or when studying.

Skill 1 — Naming a point you can see

Rule: An ordered pair is (x, y) — **across first, then up.**

x counts squares to the right of the origin; y counts squares above it. Swapping them names a different place, so “across, then up” is the whole of this skill.

Example: A table records where three points sit. Across: A 3, B 7, C 5. Up: A 6, B 2, C 9. Write the ordered pair for B .

Step 1. The pair must be written across-number first, so I read B out of the across row before I touch the up row.

Step 2. The across row gives B as 7 and the up row gives B as 2. $B = (7, 2)$

Try it: From the same two rows, write the ordered pair for C .

check: $(5, 9) = C$

Skill 2 — Moving from one point to another

Rule: Moving *right* changes only x ; moving *up* changes only y .

new x = old x + steps right new y = old y + steps up

Left and down subtract instead. Nothing you do sideways can change the up number.

Example: Point R is at $(3, 4)$. Move 5 units right and 2 units up. Where do you land?

Step 1. Each direction touches its own coordinate, so I handle the sideways move and the upward move separately instead of adding everything together.

Step 2. Across: $3 + 5 = 8$. Up: $4 + 2 = 6$. Landing point: $(8, 6)$

Try it: Point S is at $(6, 1)$. Move 2 units right and 7 units up. Where do you land?

check: $(8, 8)$

Skill 3 — Plotting points and measuring between them

Rule: To plot (x, y) , put a finger on the origin, count x squares right, then y squares up, and mark the corner where you stop.

Two points with the same y lie on a horizontal line, and the distance between them is the difference of their x -values. Two points with the same x lie on a vertical line, and the difference of their y -values is the distance.

Example: Plot $D(2, 1)$ and $E(8, 1)$. How many units apart are they?

Step 1. Both points have the same up number, so they sit on one horizontal line and I can count squares across instead of measuring diagonally.

Step 2. The across numbers are 2 and 8, so the gap is $8 - 2$. Distance: **6** units

Try it: Plot $F(4, 2)$ and $G(4, 9)$. How many units apart are they?

check: 7 units

Skill 4 — Turning a growing pattern into points

Rule: A rule of the form “start at s , change by c each step” makes ordered pairs (step number, value).

value after n steps = $s + c \times n$ (use $-$ when the quantity shrinks)

The step number is the across coordinate and the value is the up coordinate. Plotted, a constant-change rule always makes a straight line.

Example: A tree is 4 metres tall and grows 2 metres a year. Write the ordered pair for year 6.

Step 1. The year is the across number and the height is the up number, so I only have to compute the height and the pair is finished.

Step 2. Six years of growth adds $2 \times 6 = 12$ to the starting 4, giving $4 + 12$. Year 6 pair: **(6, 16)**

Try it: A tank holds 30 litres and loses 4 litres an hour. Write the ordered pair for hour 5.

check: (5, 10)